

L23b - Prim's Algorithm

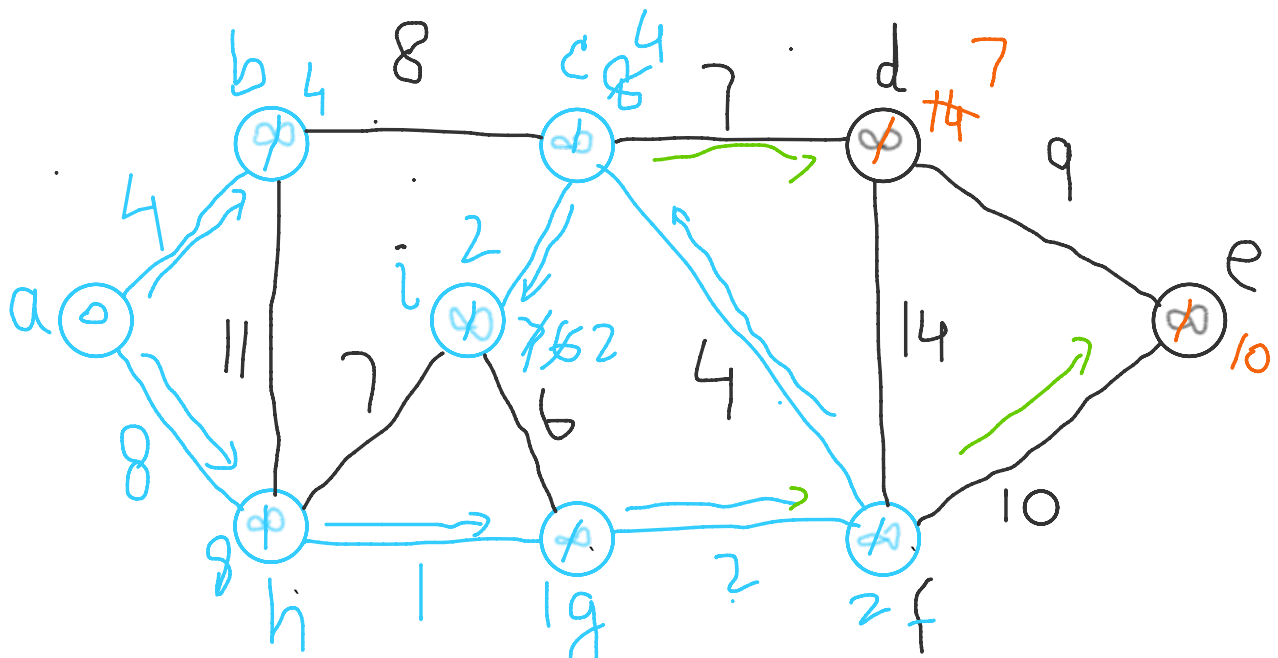
Tuesday, June 23, 2020 9:50 AM

It grows the MST vertex by vertex.

MST-PRIM(G, w, r)

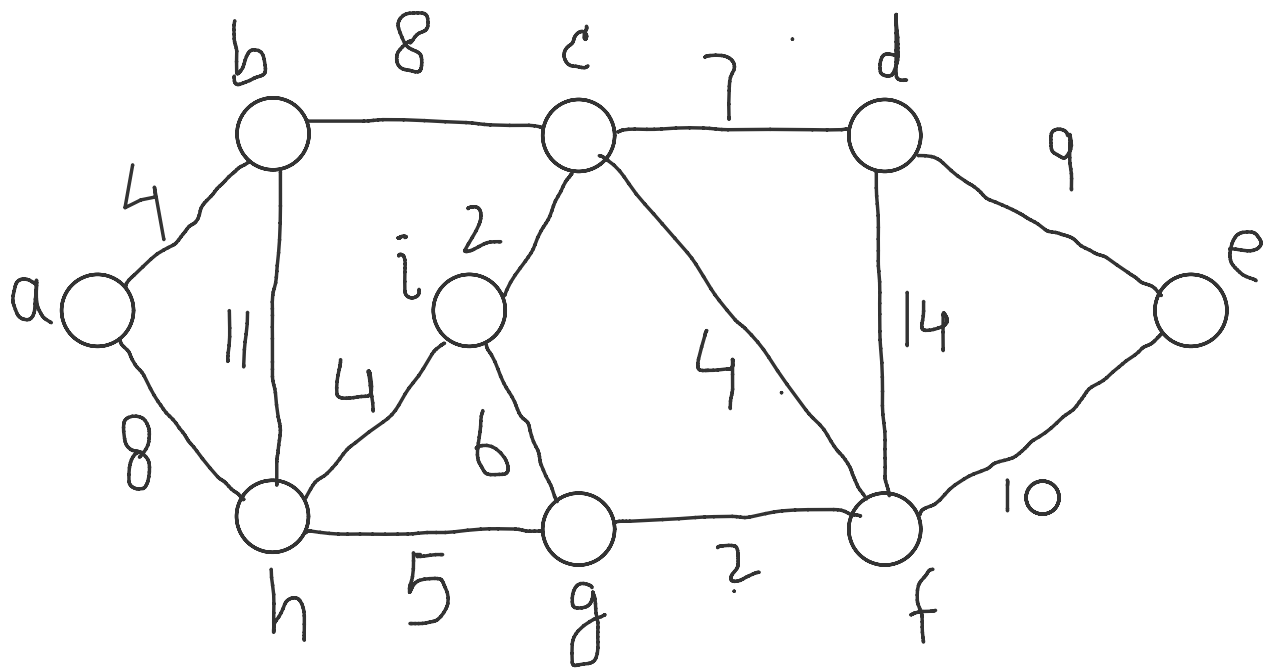
1. for each $u \in G.V$ $O(|V|)$
2. $u.key = \infty$
3. $u.\pi = NIL$
4. $r.key = 0$
5. $Q = G.V$ //Min. Priority Queue $O(|V| * \log(|V|))$ | $O(|V|)$
6. while $Q \neq \emptyset$
7. $u = \text{EXTRACT-MIN}(Q)$ $O(|V| * \log(|V|))$
8. for each $v \in G.Adj[u]$ $O(|E|)$
9. if $v \in Q$ and $w(u, v) < v.key$
10. $v.\pi = u$
11. $v.key = w(u, v)$ $O(|E| * \log(|V|))$

$$O((|E| + |V|) * \log(|V|)) = O(|E| * \log(|V|))$$



$\bar{h}$ 1 $\bar{g}$ 2 $\bar{f}$

$$4+8+2+1+2+4+7+9=37$$



$$4+8+4+2+2+4+7+9=40$$

b 8 c 7 d

